1. **Project Title:** Using gradient boosting models to combine genetic with environmental information and increase prediction ability in soybean.
2. **Principle Investigator:** Samuel B Fernandes, Assistant Professor, CSES
   Collaborators: Caio Canella Vieira, Assistant Professor, CSES
   Elvis Felipe Elli, Assistant Professor, CSES
3. **Project performance dates:**
   10/2023 - 2/2025
4. **Project Summary/Abstract**

The use of molecular markers in genomic selection offers a powerful resource to predict breeding values. However, to capture the complex interactions between genes and the environment, a model that includes environmental information is needed. In this project, we aim to improve the prediction of soybean across different environments using machine learning approaches. We will survey and organize environmental information from multiple sources and combine it with historic phenotypic information and molecular marker data currently being obtained in the University of Arkansas Soybean Breeding Program to develop predictions using machine learning models. Once we develop these prediction models, we will evaluate different training set compositions to correctly place different individuals in a sparse multi-environment trial. Ultimately, our approach could improve the prediction accuracy of GS models, which would be critical for breeding crops adaptable to unforeseen climate conditions.

5. **Project Narrative**

Introduction: Genomic selection (GS) is a powerful tool that utilizes genome-wide molecular markers to predict breeding values (Meuwissen et al., 2001). GS is expected to be superior to more traditional marker-assisted selection approaches that only consider markers in linkage disequilibrium with the largest-effect genomic signals (Lorenzana and Bernardo, 2009; Heffner et al., 2011). The combination of increasingly available dense marker data and enhanced GS models has been revolutionizing the practice of plant breeding because selections can be made on seeds for optimal predicted phenotypic performance before they are planted. Thus, one of the advantages of such models is that they make it possible to speed up breeding cycles (Watson et al., 2019). The typical GS model, which has a targeted trait as the response variable and a genome-wide set of markers as the explanatory variables, is fitted in a training population of individuals that are both phenotyped and genotyped. Selections are then made in a breeding population genotyped with the same markers based on trait values predicted by the fitted GS model (Heffner et al., 2009). Because of these benefits of GS, resources that traditionally go into obtaining phenotypic values can be reallocated elsewhere. Consequently, breeders and statisticians are rethinking several key components of plant breeding trials. For example, because markers allow the connection of alleles across replicates and environments, the optimal allocation of breeding progenies across experimental entries will likely change (Lorenz, 2013; Endelman et al., 2014). This has encouraged the development of GS models that specifically capture genotype-by-environment (G×E) interactions. Such models have been extensively tested and reported (Jarquin et al., 2014). However, the phenotypic expression results from genetic, environmental, and interaction effects. Thus, molecular markers alone may not reflect the complex interactions between genes and the environment that ultimately drive complex genotype-by-environment interactions. Therefore, our objectives are to I) organize and utilize publicly available environmental information in soybean prediction; II) deploy machine

learning approaches in designing sparse testing schemes that can be deployed in soybean; and ii) delineate alternatives to utilize crop modeling in the genomic prediction pipelines.

Rationale and Significance: Breeding crops adaptable to various conditions will be essential to maintaining a supply of food, feed, and fiber because of the changing environments that crops are facing. An effective way to predict phenotypic values under these uncertain climate conditions would be through the development of GS models that use environmental alongside genetic information. However, current methods are not well suited to predicting how specific genotypes will perform across environments (i.e., genotype-by-environment interaction). Utilizing machine learning approaches and environmental data in maize, members of the Fernandes lab were able to attain a second place in a national competition for phenotypic prediction. Here, we expand on this earlier work and hypothesize that the inclusion of variation in genetic and genotype-by-environmental information is a critical piece of information for the improvement of predicting the performance of specific genotypes across different environments in soybean. We envision that the results of this project will shed light on the placement of individuals in specific locations for multi-environment trials designed for genomic prediction and will be relevant to soybean and other crops. Ultimately, we explore ML approaches to predict phenotypic performance for specific environments that will become relevant in the future climatic stresses that agriculture faces. The development of an approach that efficiently allocates genotypes to specific environments in sparse testing would mean that not all individuals would have to be evaluated in all locations. Consequently, without increasing the budget, a given breeding program would be able to increase the number of lines being evaluated, which could result in higher selection intensity and genetic gain. Alternatively, a smaller set of phenotypic evaluations would be necessary to attain a similar rate of genetic gain.

Approach: The first stage of this project will consist of surveying and organizing environmental information from multiple sources (e.g., Web Soil Survey, NASA, and possible local weather stations). The environmental information will be combined with historic phenotypic information from the soybean breeding program to develop predictions using machine learning (e.g., gradient-boosting) models. The second stage will combine the molecular marker data set (>1,000 soybean lines) under development by the Vieira lab with the environmental data to model genotype-by-environment interaction in soybean. Machine learning models utilizing genetic and environmental data alone and combined will be evaluated using a hold-out approach. The model-based prediction will be compared to an environment (i.e., year or location) not used in the training set. Finally, after developing a prediction model, we will evaluate different training set compositions varying the number of individuals coming from different locations and identify a pattern that allows for the correct placement of different individuals in a sparse multi-environment trial. The last step to improve the prediction model will be to use the information from the multi-environmental model in defining an approach to obtain environmental covariates (with crop modeling) that can further improve genomic prediction accuracy.

Activity Description	2023				2024				2025			
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
Data survey/acquisition			X	X								
Development of environmental, genetic, and genotype-by-environment interaction prediction models					X	X	X	X				
Designing strategies to use environmental covariates									X			

6. How will this project be leveraged to apply for extramural funding?

This project will allow us to generate proof-of-concept for a NIFA proposal on validating sparse testing in soybean genomic prediction. We will develop in soybean what we partly developed in maize for prediction models. Once we show evidence that such prediction models could work in soybean, we will be able to submit a proposal that actually tests whether or not our strategy can be translated into real genetic gain.

7. Citations

Endelman, J.B., G.N. Atlin, Y. Beyene, K. Semagn, X. Zhang, M.E. Sorrells, and J.L. Jannink. 2014. Optimal design of preliminary yield trials with genome-wide markers. Crop Sci. 54(1): 48–59.

Fugeray-Scarbel, A., Bastien, C., Dupont-Nivet, M., & Lemarié, S. (2021). Why and How to Switch to Genomic Selection: Lessons From Plant and Animal Breeding Experience. Frontiers in Genetics, 12(July), 1–9.

Heffner, E.L., M.E. Sorrells, and J.-L. Jannink. 2009. Genomic selection for crop improvement. Crop Sci. 49(1): 1–12.

Jarquín, D., J. Crossa, X. Lacaze, P. Du Cheyron, J. Daucourt, J. Lorgeou, F. Piraux, L. Guerreiro, P. Pérez, and M. Calus. 2014. A reaction norm model for genomic selection using high-dimensional genomic and environmental data. Theor. Appl. Genet. 127(3): 595–607.

Lorenz, A.J. 2013. Resource allocation for maximizing prediction accuracy and genetic gain of genomic selection in plant breeding: a simulation experiment. G3 (Bethesda). 3(3): 481–491.

Lorenzana, R.E., and R. Bernardo. 2009. Accuracy of genotypic value predictions for marker-based selection in biparental plant populations. Theor. Appl. Genet. 120(1): 151–161.

Meuwissen, T. H. E., Hayes, B. J., & Goddard, M. E. 2001. Prediction of total genetic value using genome-wide dense marker maps. *Genetics*, *157*(4), 1819–1829.

Watson, A., Hickey, L. T., Christopher, J., Rutkoski, J., Poland, J., & Hayes, B. J. (2019). Multivariate genomic selection and potential of rapid indirect selection with speed breeding in spring wheat. *Crop Science*, *59*(5), 1945–1959. https://doi.org/10.2135/cropsci2018.12.0757

8. Budget

Title: Using gradient boosting models to combine genetic with environmental information and increase prediction ability in soybean. PIs: Samuel B Fernandes Project dates: 10/2023 - 2/2025	
	Budget ($)

	PI	Co-PI	Co-PI
SALARIES			
GRA(s)	25,840		
Staff			
Hourly, non-student			
Hourly, enrolled student			
FRINGES	1,160		
TRAVEL - Domestic	2,000		
TRAVEL - Foreign			
EQUIPMENT			
MATERIALS & SUPPLIES			
JOURNAL PUBLICATION FEES	1,000		
OTHER DIRECT COSTS			
TOTAL			

Budget Narrative:
The budget will be mostly allocated on an experienced graduate student ($18,340 base, $7,500 tuition, and $1,160 benefits) that will work under my supervision to obtain, process, and utilize the data in the prediction models. A total of $2,000 in out-of-state travel is requested to attend and present results in scientific meetings. We also request 1,000 cover part of the publication fees. The remaining cost will come from Dr. Fernandes' startup package.

9. Current and Pending Support

NAME (List/PD #1 first)	SUPPORTING AGENCY AND AGENCY ACTIVE AWARD/PENDING PROPOSAL NUMBER	TOTAL $ AMOUNT	EFFECTIVE AND EXPIRATION DATES	% OF TIME COMMITTED	TITLE OF PROJECT
	Active:				
M.L. Worthingon, R.T., Threlfall, S.B. Fernandes, S. Lafontaine	USDA-NIFA	~$650,000	2023-2027	5%	Integrating Sensory, Genomic, and Metabolomic Data to Breed Blackberries for Improved Flavor
	Pending:				
C.C. Vieira and S. B. Fernandes	Soybean Board	~$101,600	2023-2025	5%	Genomic Prediction to Enhance the Efficiency of

					Soybean Breeding